AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 36 · TEACHER GUIDE

# Neural network experiment defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

## Reward a bounded explanation backed by inspectable evidence.

Use checked calculations and independent case reasoning rather than presentation polish.

**Keep one habit:** Model a defense → challenge unsupported claims → assess fresh C1/C2 evidence → specify the next fair comparison.

MAIN LESSON · SECTION 1

## Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

### Target and main teaching plan

Primary target: Defend a network experiment using traceable **evidence**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

## Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Fictional data only; reserve the independent task and fresh retry until assessment

No. They may support regression checks, but a revised procedure needs fresh reserved cases for independent evaluation.

## Opening retrieval

Without opening the old lesson, explain one key idea from Capstone **limitation**s and revision. Give an example and a mistake that the idea helps you avoid.

Look for: Evaluate a frozen candidate and baseline on final rows

Reference example, not the only acceptable wording: Teacher explanation: Residuals 0,0,2,4; squared errors 0,0,4,16. The ReLU **prediction**s for both negative **input**s are zero.

Without opening the old lesson, explain one key idea from Capstone hypothesis and baseline. Give an example and a mistake that the idea helps you avoid.

Look for: Write a testable capstone hypothesis and a credible baseline

Reference example, not the only acceptable wording: Teacher explanation: Predict max(0, a*x) without bias; baseline a = 1; candidate starts at 1, learning rate 0.1, two full-batch updates on T1/T2.

Hypothesis: candidate validation MSE on V1/V2 is strictly below the baseline MSE.

Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

Look for: Explain how a changed **input** mix can change overall accuracy

Reference example, not the only acceptable wording: Teacher explanation: Clear accuracy is 90 percent in both cohorts; dim accuracy is 50 percent in both.

Development has 80 percent clear and 20 percent dim; new use has 20 percent clear and 80 percent dim. The mix changes while observed group rates remain the same.

## Pictured end goal

Explain neural network experiment **defense** with a traceable toy-network case and a bounded conclusion.

## ▼ Earlier wording, alternatives and picture transcripts

### Teach the learning cycle

### Retrieval answer guidance

### Capstone limitations and revision

### Capstone hypothesis and baseline

### Dataset shift in neural models

For the compact alternative schedule, use two 45-minute sessions. This capstone uses constructed scalar ReLU regression so every calculation is inspectable. It is an experiment in **training** and **evidence**, not a real-world deployment.

Preserve written plans, data roles and raw **prediction**s. Optional local Python requires no external packages or AI account. Do not expose independent-task solutions before collecting work.

### Outcomes

Defend a network experiment using traceable **evidence**

Separate selection, final evaluation and revision claims

Answer a challenge with a calculation and a bounded conclusion

### Preparation

### Prerequisite answer

### End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

# Assess the explanation rather than the polish of the demonstration

> Require traceable calculations, data roles and appropriately limited claims.

## Main inventory, selection and complete frozen errors

A **defense** is not a sales pitch. Assemble the original question and criterion, data roles, baseline, model equation, settings, raw checkpoints, validation selection, frozen final **prediction**s, **limitation**s and revision history.

Someone reading the record should be able to recompute a reported MSE and see which decisions each dataset informed.

For this capstone, two rates were compared under matched conditions. Rate 0.2 selected a = 2 with validation MSE 0 versus 0.78125 for rate 0.1.

The frozen candidate then achieved final MSE 5 versus baseline 20.25 on four constructed cases. These are different comparisons and must not be collapsed into “the network was 100% accurate”.

| Stage | Evidence | Supported statement |
|---|---|---|
| Validation selection | Two positive rows; MSE 0 versus 0.78125 | Selected rate 0.2 by written rule |
| Frozen final comparison | Four rows; MSE 5 versus 20.25 | Lower MSE than initial-weight baseline on these rows |
| Condition check | Two negative rows; candidate MSE 10 | Nonnegative output fails to represent negative targets |
| Proposed revision | Signed output on revealed cases | A possible representational repair, not new final evidence |

Final ID · x · Target · Baseline prediction · Candidate prediction · Baseline squared error · Candidate squared error

> **Final ID:** F1
>
> **x:** 5

**Target:** 10

**Baseline prediction:** 5

**Candidate prediction:** 10

**Baseline squared error:** 25

**Candidate squared error:** 0

**Final ID:** F2

**x:** 6

**Target:** 12

**Baseline prediction:** 6

**Candidate prediction:** 12

**Baseline squared error:** 36

**Candidate squared error:** 0

**Final ID:** F3

**x:** -1

**Target:** -2

**Baseline prediction:** 0

**Candidate prediction:** 0

**Baseline squared error:** 4

**Candidate squared error:** 4

**Final ID:** F4

**x:** -2

**Target:** -4

**Baseline prediction:** 0

**Candidate prediction:** 0

**Baseline squared error:** 16

**Candidate squared error:** 16

> **Final ID:** Four-row MSE
>
> **x:** Not classification accuracy
>
> **Target:** Same cases
>
> **Baseline prediction:** 20.25
>
> **Candidate prediction:** 5
>
> **Baseline squared error:** Negative candidate subset
>
> **Candidate squared error:** 10 on two rows

## Supported reduction key and main challenge answers

### Keep the evidence roles separate

| PLAN OR INPUT | PROCEDURE | PERMITTED CONCLUSION |
| --- | --- | --- |
| Question and settings | Replay calculations | Procedure can be checked |
| Frozen final predictions | Compare same four rows | Bounded baseline improvement |
| Known failures and changes | Preserve old results | Revision still needs evaluation |

Every claim has an evidence source and a limit. A complete record is not a substitute for broader data.

Guided: 8 - 2 = 6 MSE units; 6/8 = 75% relative MSE reduction on four final rows. It is not classification accuracy.

Exit: validation MSE 0 on two rows selected the candidate; frozen final MSE 5 versus 20.25 on four rows, with negative-**output** failures. Timing, stronger baselines and independent revised-model evaluation are absent.

Challenge: “Your validation error was zero, so why did the final test fail on negative targets?” Response: validation contained only positive **input**s; the **output** max(0,a*x) cannot be negative.

The final negative cases expose a coverage and architecture **limitation**. The overall criterion still passed, but that does not remove the failure.

Challenge: “Is a neural network better than linear regression?” Response: the supplied baseline was a fixed initial weight, not a fitted line. The **evidence** establishes improvement relative to that baseline only.

A fitted signed line is a relevant additional comparison and needs a documented, fair evaluation. Challenge: “Was the candidate faster?” Response: no timing **evidence** was collected, so speed is unmeasured.

## Separate pictured neuron and numerical scope

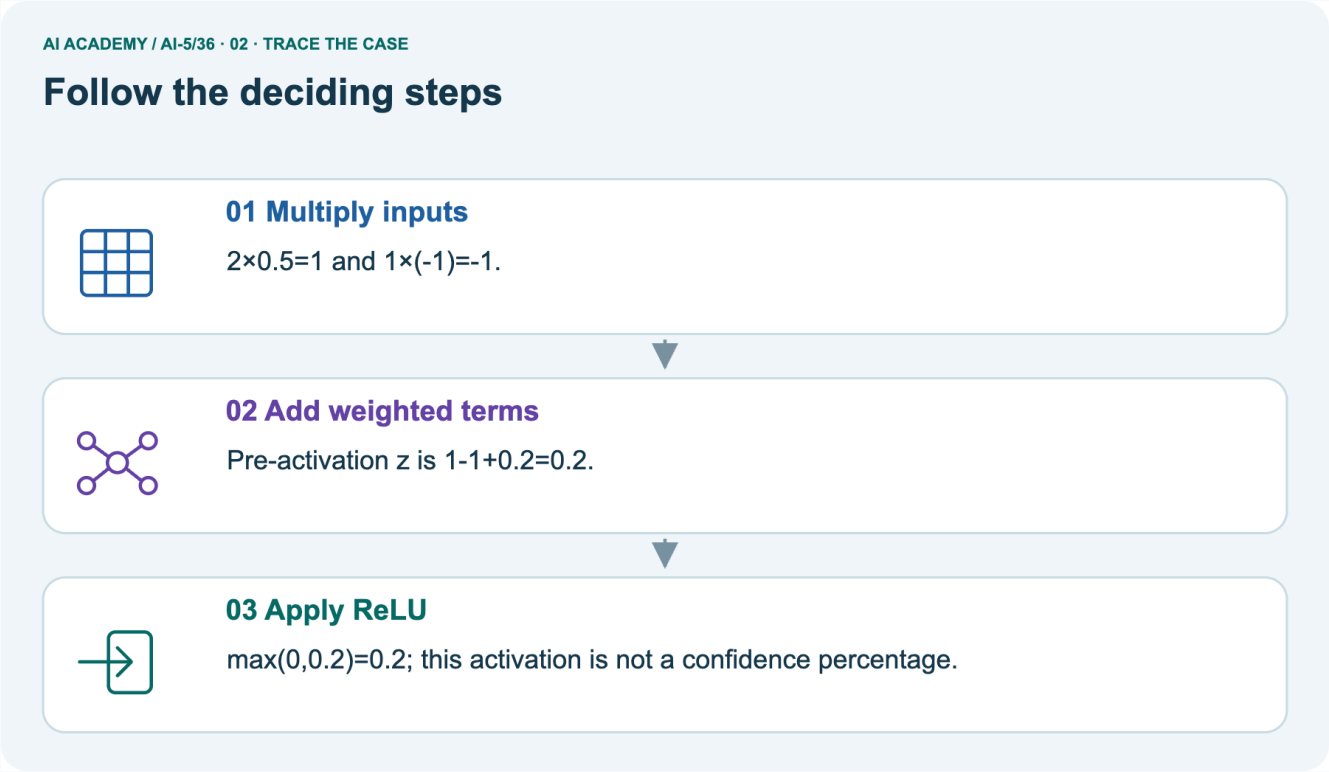
Defend an individual network calculation. At the defence, a team shows a single toy neuron before describing the whole image-pattern experiment.

**Input**s are [2,1]; weights [0.5,-1]; bias 0.2; activation is ReLU, max(0,z). This is a demonstration, not a probability model.

The picture is a separate arithmetic demonstration, not the scalar capstone’s evaluation. Inputs [2, 1], weights [.5, –1], bias .2 give weighted terms 1 and –1, preactivation .2 and ReLU output .2. Changing only the second input to 2 gives z = –.8 and output 0. There is no supplied class-reference dataset, probability conversion or accuracy denominator for this neuron.

Its output .2 cannot be called 20% confidence or 20% accuracy.

## Pictured calculation and changed-input key




Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Multiply inputs | 2×0.5=1 and 1×(-1)=-1. |
| 2. Add weighted terms | Pre-activation z is 1-1+0.2=0.2. |
| 3. Apply ReLU | max(0,0.2)=0.2; this activation is not a confidence percentage. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A defence should distinguish a traceable network calculation from a claim about classification quality. The neuron transforms supplied numbers; held-out labelled cases are needed to assess the complete system.

This one-neuron arithmetic does not establish how a trained image classifier generalises or why its full network makes a decision.

Change: Change the second **input** from 1 to 2.

**Prediction**: z becomes 1-2+0.2=-0.8 and ReLU returns 0.

Reason: The negative weighted term grows; ReLU clips a negative pre-activation to zero.

### ▼ Earlier wording, alternatives and picture transcripts

| Term | Meaning |
| --- | --- |
| Defense | An explanation of methods and claims supported by inspectable evidence |
| Evidence chain | The connection from inputs and procedure through raw outputs to a reported claim |

| Term | Meaning |
| --- | --- |
| Limitation | A condition restricting what evidence can establish |
| Reproduction | Repeating the documented procedure to check reported outputs |

## Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

| Thinking move | Teacher prompt |
| --- | --- |
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

## A pictorial worked case: Neural network experiment defense

### The facts we will use

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#### One toy ReLU neuron: not a probability

**Input 2 x weight 0.5 = 1**

**Input 1 x weight -1 = -1**

Add the products and bias 0.2: total = 0.2

**ReLU: use max(0, total) -> output 0.2**

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/36

# Neural network experiment defense

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Multiply inputs

2 x 0.5=1 and 1 x (-1)=-1.



## 2. Add weighted terms

Pre-activation z is 1-1+0.2=0.2.



## 3. Apply ReLU

max(0,0.2)=0.2; this activation is not a confidence percentage.

**CHANGE ONE THING**

Change the second input from 1 to 2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

**Why this result follows**

**What this example does not establish**

**Change one thing in the pictured case**

**Ask while showing the picture**

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

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MAIN LESSON · SECTION 3

# Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

## Current completion task and checked reasoning

Completion task: State the validation selection and frozen final comparison using their correct metrics and scopes.

Teacher check: Teacher explanation: Rate 0.2 selected a = 2 with validation MSE 0 versus 0.78125 on two positive rows.

Frozen final MSE was 5 versus baseline 20.25 on four constructed rows; negative-**input** candidate MSE was 10 on two rows.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

## Individual misconception check

Use the worked case for Neural network experiment **defense**. Proposed explanation: “A polished demonstration is an evaluation”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A demonstration needs traceable data roles, metrics and raw results to support a performance claim.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

## Pictured reasoning and changed case

Calculate both weighted terms, z and ReLU **output**. Predict the changed second **input** and explain why 0.2 cannot be called 20% accuracy.

The products are 1 and -1; z=0.2 and ReLU=0.2. With second **input** 2, z=-0.8 and **output**=0. The activation is a neuron value, not an accuracy fraction or calibrated probability.

Changed-case reasoning: z becomes 1-2+0.2=-0.8 and ReLU returns 0. The negative weighted term grows; ReLU clips a negative pre-activation to zero.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

## Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a frame with question, fixed settings, changed setting, data roles, raw **prediction**s, metric, conclusion and **limitation**. Allow a calculator and explicit squared-error columns.

Extension: implement the arithmetic locally, preserve the program version, and explain why a successful replay adds no independent test cases.

## ▼ Earlier wording, alternatives and picture transcripts

| Time or stage | Teacher action |
|---|---|
| Session 1 start to 7 minutes | Retrieve the prerequisite and ask students to explain the data roles. |
| 7 to 18 minutes | Read section 1 and complete W1 with a recorded reason. |
| 18 to 32 minutes | Trace section 2 and complete W2; compare raw values before conclusions. |
| 32 to 45 minutes | Explain the diagram and complete W3 in pairs; record any support. |
| Session 2 start to 8 minutes | Complete the guided response and discuss why the evidence has limits. |
| 8 to 20 minutes | Read the remaining explanation and prepare a concise experiment record. |
| 20 to 35 minutes | Collect W4 independently. Require calculations and a bounded claim. |
| 35 to 45 minutes | Use W5 and the exit response for feedback. Administer the fresh retry after feedback without the earlier solution. |

### Reduce help deliberately

### Individual reasoning check

### Pictorial practice question

### Reasoned teacher answer

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MAIN LESSON · SECTION 4

# Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

## Complete inputs,expected results and limits

Main selection validation 0 versus.78125 on two rows; frozen final candidate 5 versus 20.25 on four, negative subset 10. Supported8→2 gives 6 units/75% relative MSE reduction. Independent selects C2 from validation 1.5/1.0. Baseline final squares 4, 9, 1, 2 sum 16→MSE 4; C2 squares 1, 1, 2, 0 sum 4→MSE 1; reduction 3 units/75%. Last-two-row condition B baseline 1.5 versus candidate 1, but its first row worsens 1→2.

No speed, energy or universal reliability claim supported.

## Diagnose the current misconceptions

A polished demonstration is an evaluation: A demonstration needs traceable data roles, metrics and raw results to support a performance claim.

Relative MSE reduction is accuracy: Regression MSE and classification accuracy are different metrics.

A **limitation** invalidates every result: A **limitation** bounds a result; lower error on the measured rows can be true without a universal claim.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

## Choose accessible support

| What the response shows | Next teaching action |
| --- | --- |
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

**Match the support to the first error**

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MAIN LESSON · SECTION 5

## Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

### Selected independent task and checked answer

Selected case: A separate experiment compares candidates C1 and C2 on two validation rows: MSE 1.5 and 1.0. The **written rule** selects lower MSE.

After freezing C2, four fresh final rows give baseline squared errors 4,9,1,2 and C2 squared errors 1,1,2,0. Compute final MSE and relative reduction.

Two condition-B rows are the last two rows; compare baseline and candidate MSE there. State what this record supports and what is unmeasured.

Teacher explanation: Select C2. Baseline final MSE 16/4 = 4; candidate MSE 4/4 = 1; absolute reduction 3 and relative reduction 75%.

Condition B baseline MSE (1+2)/2 = 1.5 and candidate MSE (2+0)/2 = 1, though one condition-B row worsened from 1 to 2.

The record supports lower MSE on these four final rows, not universal reliability or a classification-accuracy percentage. Speed and energy are unmeasured.

## Judge explanation,trace and limits

| Dimension | Evidence required |
| --- | --- |
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

## Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

## W1 Build the evidence inventory

Name eight items needed to check the capstone from question through final result.

Teacher explanation: Question/criterion, data and split IDs, baseline rule, architecture, initial values and learning rates, update count/loss, checkpoints and raw **prediction**s, validation selection rule, frozen final record, **limitation**s and revision history are acceptable items.

## W2 Defend the result

State the validation selection and frozen final comparison using their correct metrics and scopes.

Teacher explanation: Rate 0.2 selected a = 2 with validation MSE 0 versus 0.78125 on two positive rows. Frozen final MSE was 5 versus baseline 20.25 on four constructed rows; negative-**input** candidate MSE was 10 on two rows.

## W3 Challenge an unsupported claim

A presenter claims the experiment proves neural networks beat fitted lines and use less energy. What **evidence** is missing?

Teacher explanation: No fitted-line comparison or energy measurement was supplied.

The initial-weight baseline comparison supports neither claim; specify a fair fitted-line procedure and measured resource boundary before comparing.

## W5 Record the next investigation

Propose a fair next comparison with a fitted signed line. State how to preserve previous **evidence** and reserve new evaluation.

Teacher explanation: Keep the original plans, **output**s and failures.

Define and fit the signed-line baseline with development data, match task and evaluation metric, freeze both procedures, and use fresh reserved cases with positive and negative **input** coverage.

Record revised settings and report all outcomes.

### ▼ Earlier wording, alternatives and picture transcripts

#### Guided and exit questions

Guided: 8 - 2 = 6 MSE units; 6/8 = 75% relative MSE reduction on four final rows. It is not classification accuracy.

Exit: validation MSE 0 on two rows selected the candidate; frozen final MSE 5 versus 20.25 on four rows, with negative-**output** failures. Timing, stronger baselines and independent revised-model evaluation are absent.

### W1 Build the evidence inventory

### W2 Defend the result

### W3 Challenge an unsupported claim

### W4 Defend a new record independently

A separate experiment compares candidates C1 and C2 on two validation rows: MSE 1.5 and 1.0. The **written rule** selects lower MSE.

After freezing C2, four fresh final rows give baseline squared errors 4,9,1,2 and C2 squared errors 1,1,2,0. Compute final MSE and relative reduction.

Two condition-B rows are the last two rows; compare baseline and candidate MSE there. State what this record supports and what is unmeasured.

Teacher explanation: Select C2. Baseline final MSE 16/4 = 4; candidate MSE 4/4 = 1; absolute reduction 3 and relative reduction 75%.

Condition B baseline MSE (1+2)/2 = 1.5 and candidate MSE (2+0)/2 = 1, though one condition-B row worsened from 1 to 2.

The record supports lower MSE on these four final rows, not universal reliability or a classification-accuracy percentage. Speed and energy are unmeasured.

### W5 Record the next investigation

### Independent assessment and fresh retry

W4 requires selection C2, MSE 4 and 1, 75% relative reduction, condition-B MSE 1.5 and 1 and recognition of the worsened row. Fresh retry: baseline squared errors 9,9,1,1; selected candidate errors 1,1,0,2.

MSE 5 versus 1, relative reduction **80%**; condition-B MSE both 1 with one worsened row. Require a bounded four-row conclusion.

### Assess independent transfer

### Selected case answer

### Feedback that the learner uses

# Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

## Project and next-week bridge

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: A separate experiment compares candidates C1 and C2 on two validation rows: MSE 1.5 and 1.0.

The **written rule** selects lower MSE. After freezing C2, four fresh final rows give baseline squared errors 4,9,1,2 and C2 squared errors 1,1,2,0. Compute final MSE and relative reduction.

Two condition-B rows are the last two rows; compare baseline and candidate MSE there. State what this record supports and what is unmeasured.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Provide a frame with question, fixed settings, changed setting, data roles, raw **prediction**s, metric, conclusion and **limitation**. Allow a calculator and explicit squared-error columns.

Extension: implement the arithmetic locally, preserve the program version, and explain why a successful replay adds no independent test cases.

AI 5 begins generative and predictive objectives. Check understanding of distributions and vector similarity before advancing; provide a targeted retrieval activity where **evidence** is incomplete.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

## Delayed return and support record

Suggested return points in this level: plan a review after the level ends. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
|---|---|
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### ▼ Earlier wording, alternatives and picture transcripts

### Plan the delayed return

### Project evidence from this case

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

## Reference · Alternative planning checklist

> Use alongside the main route when helpful.

### Planning,assessment and project reminders

Teaching target: Defend a network experiment using traceable **evidence**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

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> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: State the validation selection and frozen final comparison using their correct metrics and scopes.

Reasoned practice answer: Teacher explanation: Rate 0.2 selected a = 2 with validation MSE 0 versus 0.78125 on two positive rows.

Frozen final MSE was 5 versus baseline 20.25 on four constructed rows; negative-**input** candidate MSE was 10 on two rows.

Check every learner before discussion: Use the worked case for Neural network experiment **defense**. Proposed explanation: “A polished demonstration is an evaluation”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A demonstration needs traceable data roles, metrics and raw results to support a performance claim.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change the second **input** from 1 to 2. Require a **prediction** and a reason before comparing. Expected result: z becomes 1-2+0.2=-0.8 and ReLU returns 0.

Reason: The negative weighted term grows; ReLU clips a negative pre-activation to zero.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

# Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

## Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and **limitation**s. Progress ticks and quiz selections alone do not establish mastery.

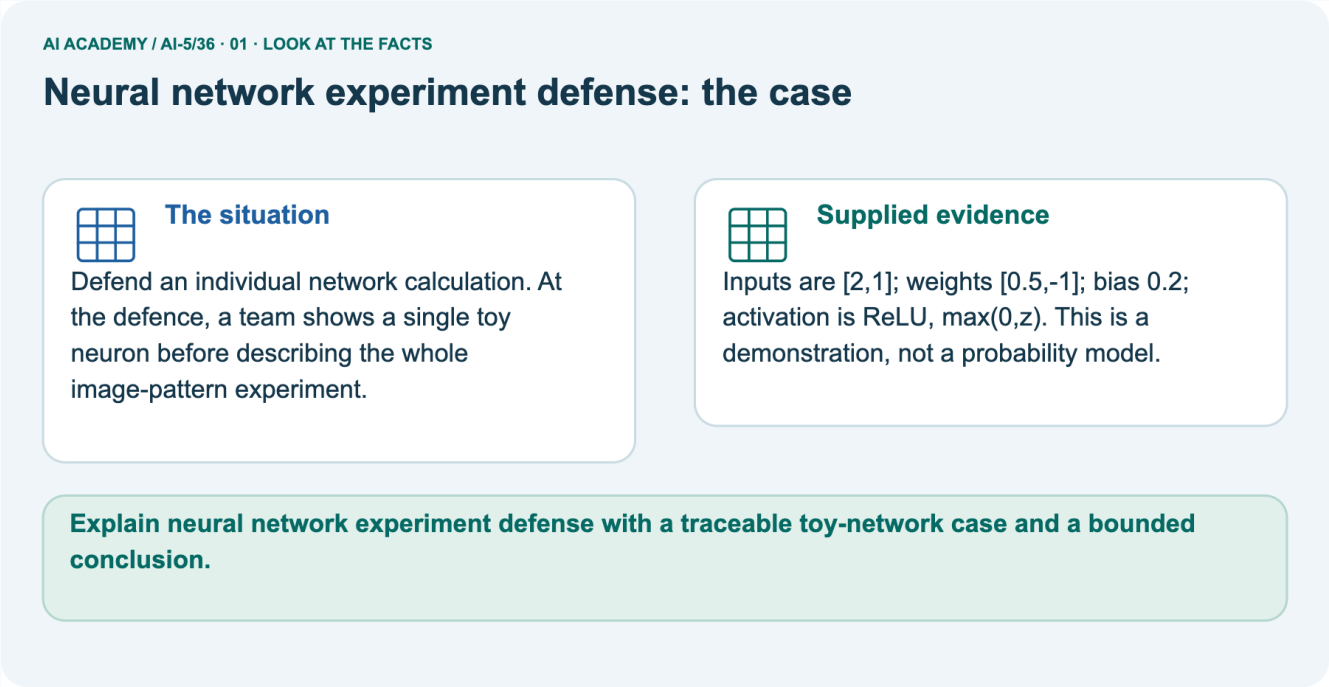
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

## Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

## 01. Neural network experiment defense: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

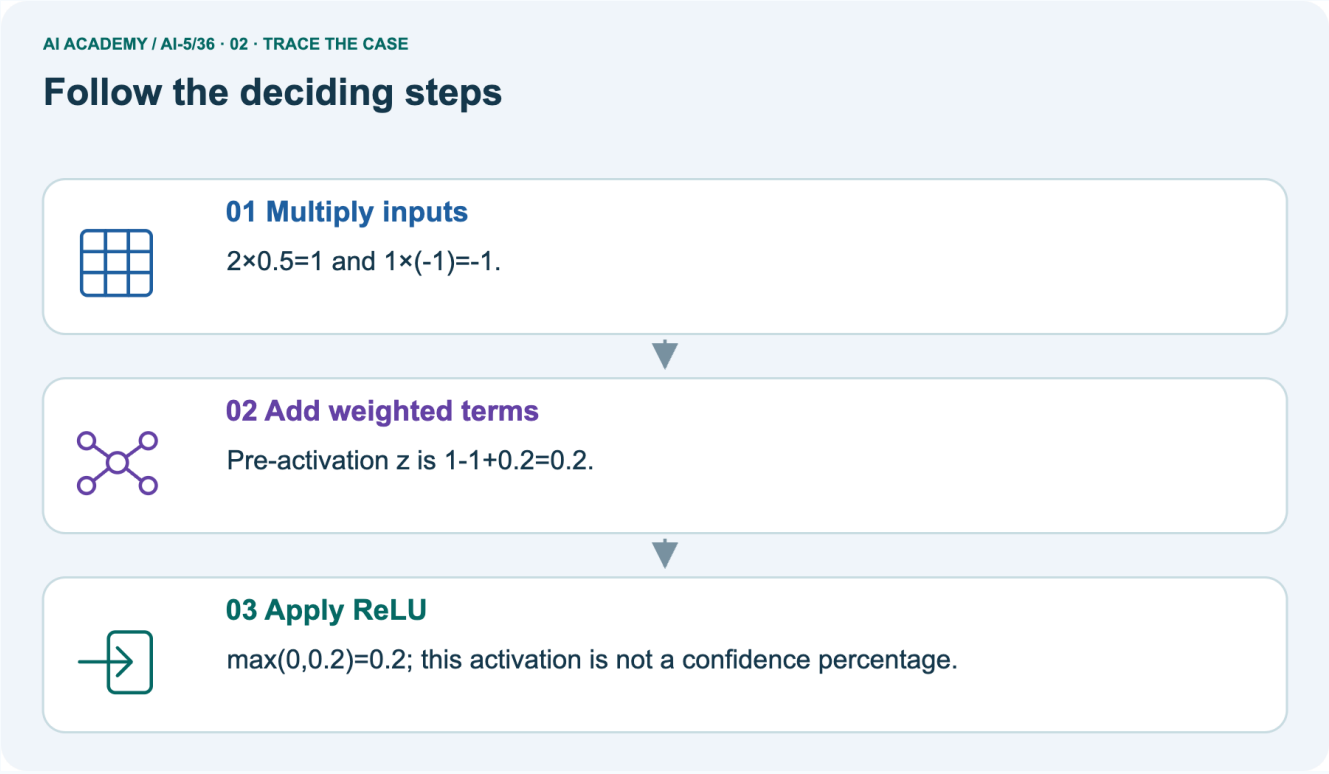
Defend an individual network calculation. At the defence, a team shows a single toy neuron before describing the whole image-pattern experiment.

Supplied facts: **Input**s are [2,1]; weights [0.5,-1]; bias 0.2; activation is ReLU, max(0,z). This is a demonstration, not a probability model.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps



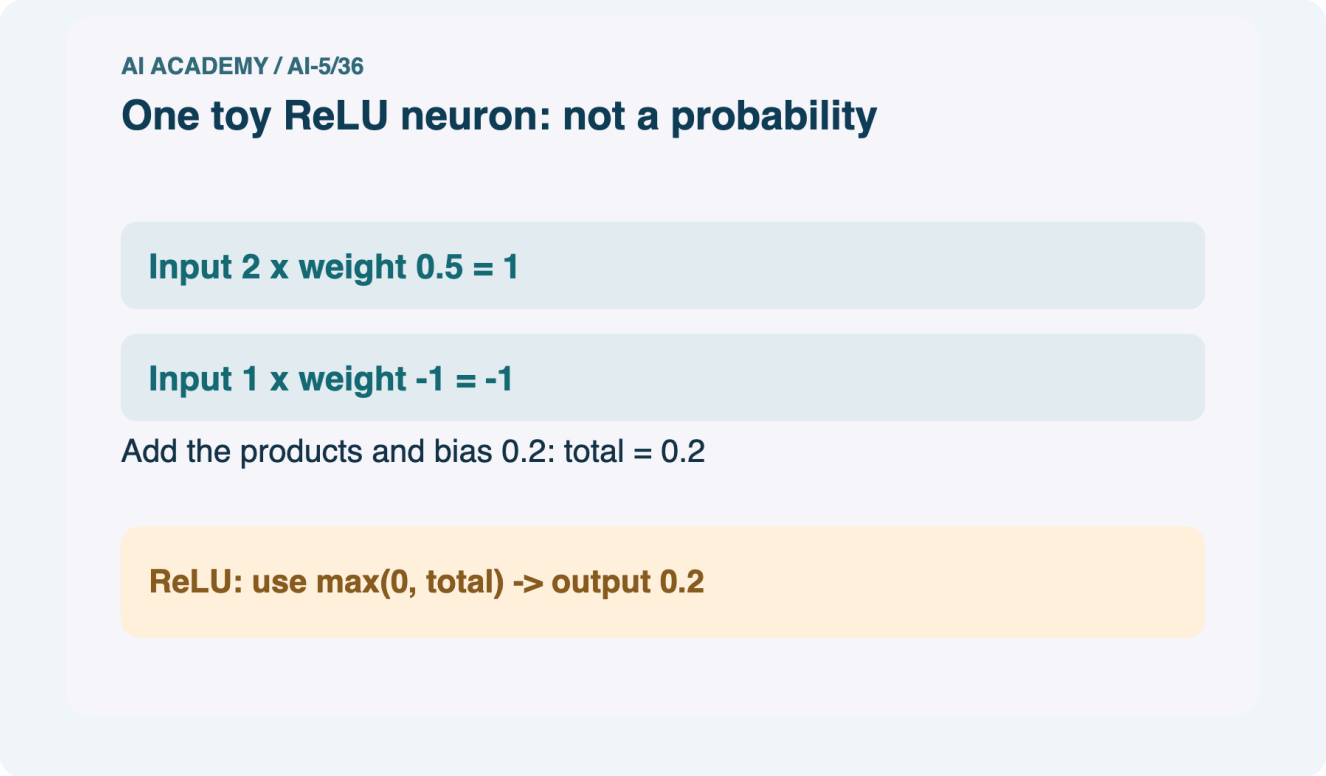

Follow the labelled picture with the matching explanation. Open the vector diagram

Multiply **input**s: 2×0.5=1 and 1×(-1)=-1. → Add weighted terms: Pre-activation z is 1-1+0.2=0.2. → Apply ReLU: max(0,0.2)=0.2; this activation is not a confidence percentage.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 03. One toy ReLU neuron: not a probability




Follow the labelled picture with the matching explanation. Open the vector diagram

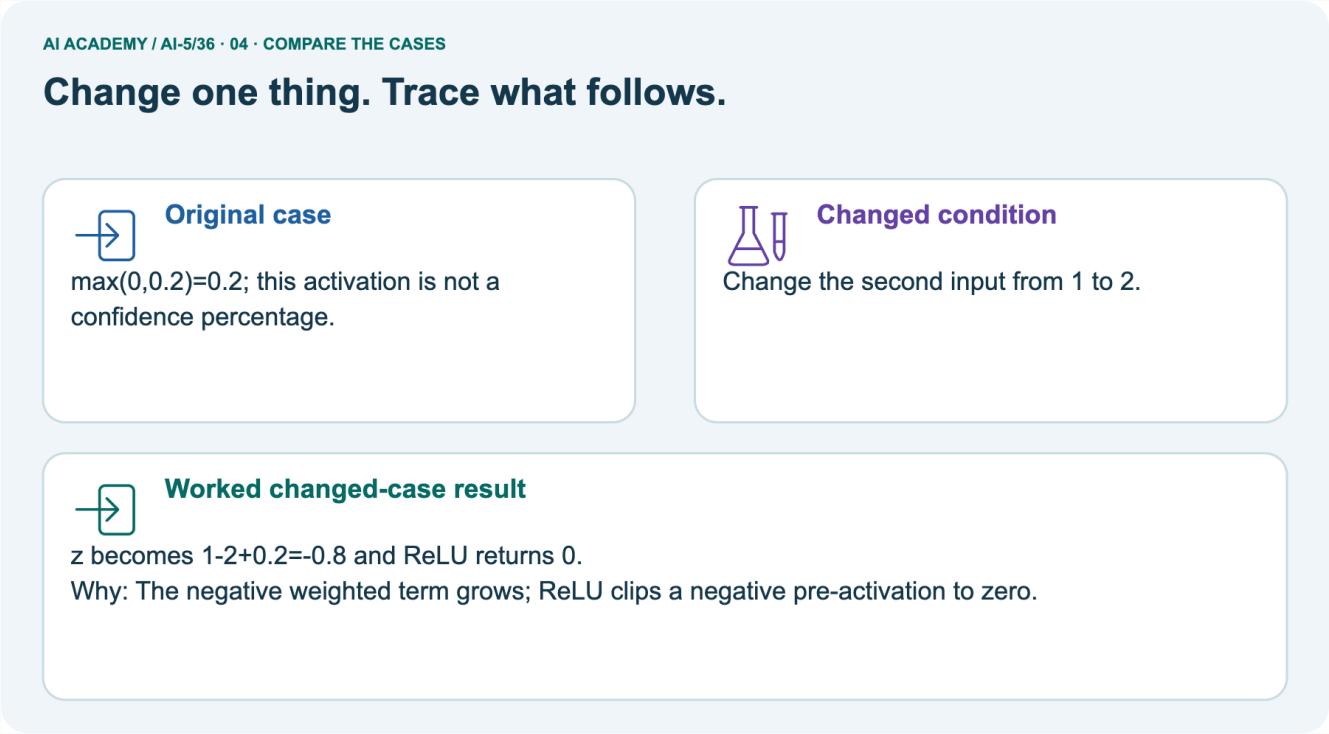
One toy ReLU neuron: not a probability. **Input**s: [2, 1]; weights: [0.5, -1]; bias: 0.2; activation: relu; **output**: 0.2.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: max(0,0.2)=0.2; this activation is not a confidence percentage. Changed condition: Change the second **input** from 1 to 2. Expected result: z becomes 1-2+0.2=-0.8 and ReLU returns 0.

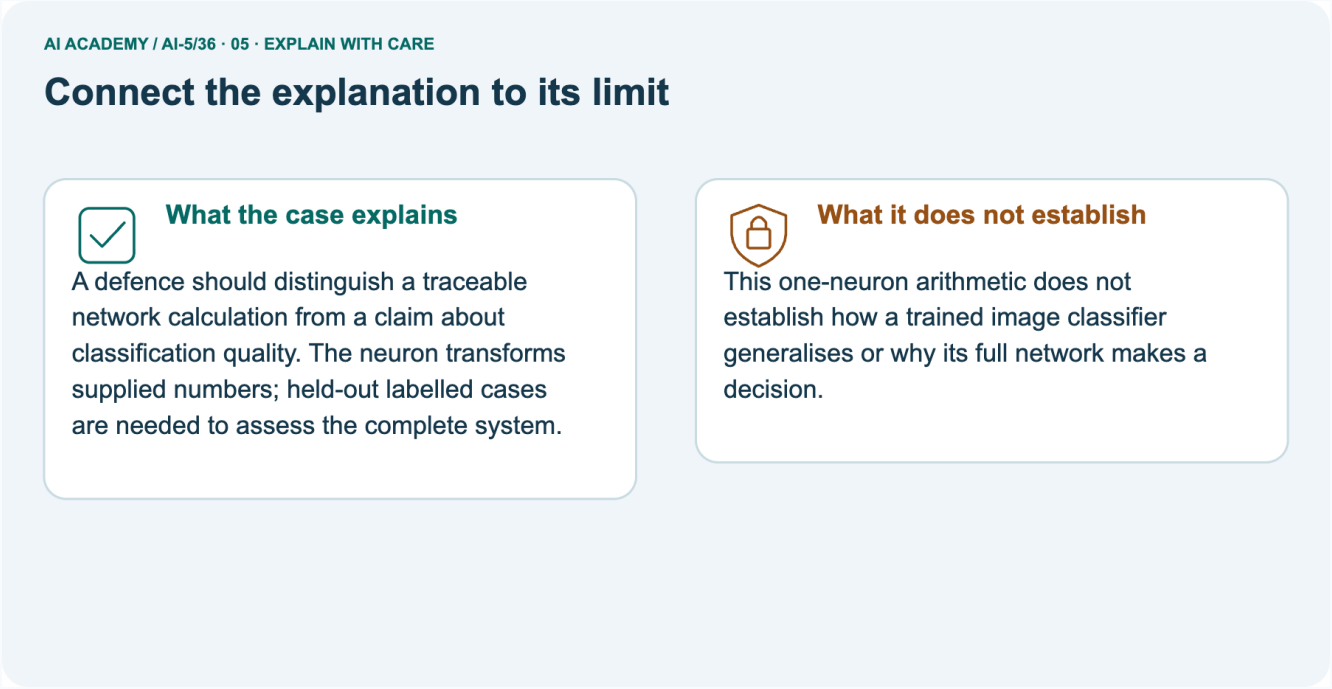
Reason: The negative weighted term grows; ReLU clips a negative pre-activation to zero.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The negative weighted term grows; ReLU clips a negative pre-activation to zero.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

A defence should distinguish a traceable network calculation from a claim about classification quality. The neuron transforms supplied numbers; held-out labelled cases are needed to assess the complete system.

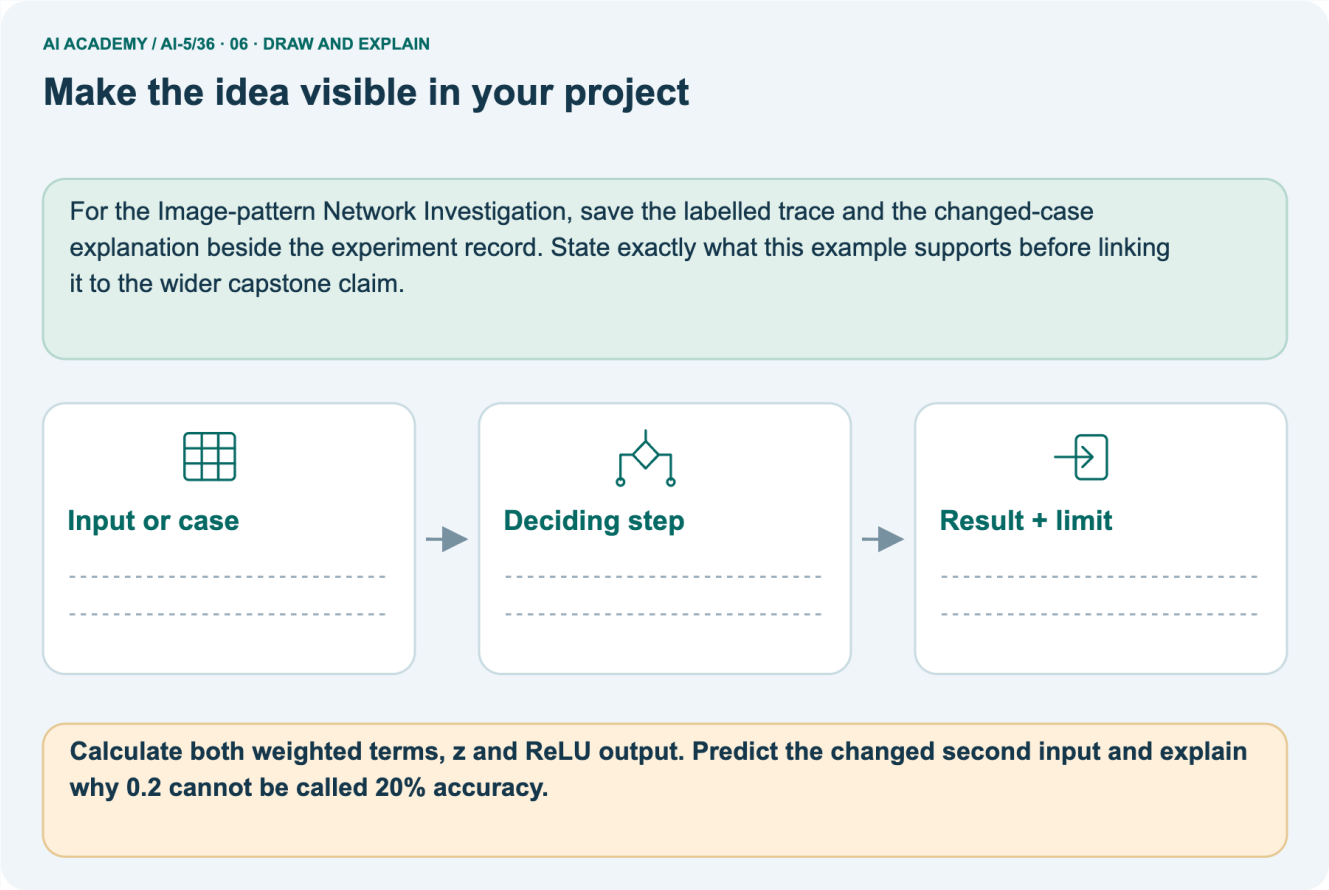
Limit: This one-neuron arithmetic does not establish how a trained image classifier generalises or why its full network makes a decision.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This one-neuron arithmetic does not establish how a trained image classifier generalises or why its full network makes a decision.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Calculate both weighted terms, z and ReLU **output**. Predict the changed second **input** and explain why 0.2 cannot be called 20% accuracy.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The products are 1 and -1; z=0.2 and ReLU=0.2. With second **input** 2, z=-0.8 and **output**=0. The activation is a neuron value, not an accuracy fraction or calibrated probability.

### ▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

**01. Neural network experiment defense: the case**

**02. Follow the deciding steps**

**03. One toy ReLU neuron: not a probability**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

### ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.